

ANTINOCICEPTIVE EFFECTS OF POLY HERBAL OIL EXTRACT (PHOE)

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ABSTRACT

A number of drugs are used widely in the traditional system of medicine or Tibb-e-Unani (Unani medicine) in the management of many diseases but these drug mostly, have not been investigated for their described effects. This study aimed to investigate antinociceptive and anti-inflammatory effects of Herbal Oil (1, 5 and 10 ml/kg) administered orally in mice and rats, using standard behavioral paradigms.

It is a poly herbal preparation used in traditional medicine as a remedy for pains and other ailments. The data show that orally administered herbal oil promotes antinociceptive activity against Acetic Acid induced writhing and Formalin induced pain models. The oil also produced a dose dependent inhibition of carrageenin-induced rat paw edema. These effects were significant when compared with saline control group. The results indicate potent antinociceptive component in oil, which justify at least partially the folkloric use of oil in relieving pains.

INTRODUCTION

Pain is “an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage” (Merskey, 1986) and a very common phenomenon. There is no doubt that pain acts as a warning signal against disturbances either in the body or in the external environment of an individual. The principal objective of the treatment of pain is to remove or abolish the cause of pain. But it is not always possible to do so; hence, analgesics are used for the symptomatic treatment of pain like NSAID (Abbott *et al.*, 1996), Opioids and antidepressant (lung *et al.*, 1997). Almost all pharmacological treatments may produce side effects (Goldberg, 1984). Although alternative treatments are increasingly being used to alleviate affective disorders (Wiklund *et al.*, 1999). Alternative medicine claims to possess many safe and effective drugs useful in various disorders. Only few have been investigated for their effects. Antinociceptive herbal Oil is a poly

herbal oily preparation containing 7 different ingredients (Table 1). It is standardized preparation used as analgesic oil traditionally. Although some of its ingredients have been studied for various pharmacological effects and their reported activities are as follows:

Linum usitatissimum L. belongs to family Linaceae. In traditional medicine *Linum usitatissimum* L. used as a mild laxative and its topical application for local inflammation (Carlson, 1998). It is still official in the Chinese pharmacopeia for constipation and dry itching skin (Tu, 1992). They are used for inflammations, abscesses and relief of pain in American conventional medicine (Taber, 1962).

Trachyspermum ammi (L.) Sprague belongs to family Umbelliferae/Apiaceae. According to Azerbaijani folk medicine, decoctions of ajowan seeds have analgesic properties (Iravani, 1834; Mu'min, 1713 and Ansari, 1818).

Table-1
Ingredients of Herbal oil

Unani Name	Scientific Name	Parts used
Alsi, Tukme Katan	<i>Linum usitatissimum</i> Linn.	Seeds oil
Ajwain Desi	<i>Trachyspermium ammi</i> (L) Sprague	Seeds oil
Jawatri	<i>Myristica fragrans</i> Houtt.	Seed
Long	<i>Syzygium aromaticum</i> (Linn.)	Flower buds
Suranjan Talkh	<i>Colchicum luteum</i> Baker	Roots or Tuber
Mal Kangni	<i>Celastrus paniculata</i> Willd	Seeds
Behroza	<i>Pinus roxburghii</i> Sargent.	Oleo-resin

Myristica fragrans Houtt. belongs to family Myristacaceae, has been found to exhibit anti-inflammatory activity (Ozaki *et al.*, 1989). Hussain and Rao (1991) reported the chemopreventive action of *Myristica fragrans* on 3-methylcholantbrene (MCA)-induced carcinogenesis in the uterine cervix of virgin mice.

Syzygium aromaticum (Linn.) belongs to family Myrtaceae. Antioxidant potential of the *syzygium aromaticum* was reported by Shymala *et al.* (2003), suggest that it possesses the hypolipidemic effect in rats may be due to its ability to combat oxidative stress by quenching free radicals generated in the body.

Colchicum luteum Baker belongs to family Liliaceae is antirhyrmatic and to allay pain in arthritis and gout (Khan *et al.*, 1997).

Celastris paniculata (Willd.) belongs to family Liliaceae. The extract showed marked CNS depressing activity (Singh *et al.*, 1974). It is also considered to be analgesic (Duke *et al.*, 1985) and used in the treatment of rheumatism, leprosy, gout, fevers and paralysis (Chopra, 1986).

Pinus roxburghii Sargent belongs to family Pinaceae. Owing to counter-irritant and anti-inflammatory properties of the pine needle baths, they may relieve allergic itch (Damirov, 1988). Pine oil has strong Anti-inflammatory, antiseptic and diuretic

properties, it also promotes granulation of wounds and is used as disinfectant and deodorant (Goryachev, 1952).

Keeping this view in mind about the pharmacological effects of different plants a herbal oil is prepared and the present study was conducted to determine the medicinal properties of herbal oil, we investigated the analgesic and anti-inflammatory activities using different animals models.

MATERIALS AND METHODS

Preparation of oil:

Fresh Ajwain Desi (*Trachyspermum ammi* (L.) Sprague), Jawatri (*Myristica fragrans* Houtt.), Long (*Syzygium aromaticum* Linn.) Suranjan Talkh (*Colchicum luteum* Baker), Mal Kangni (*Celastrus paniculata* Willd.) were purchased from local market, crushed and soaked in 1 lit water for 24h. Next day Alsi oil (*Linum usitatissimum* Linn) is added in this and boil it until all water is evaporated and all the ingredients have burnt than filter it and add pine oil (*Pinus roxburghii* Sargent.) in it. Keep it in sunrays for 10-15 days. At the end yellowish brown color oil was obtained and used during the whole study (Table 1).

Animals:

Adult male NMRI mice (20-25g) and Sprague Dawley rats (200-225g) were obtained from Dr. Hafiz Muhammad Ilyas Institute of Pharmacology and Herbal Sciences

Table-2
Effect of PHOE on acetic acid-induced writhing in mice

Group	5 min	10 min	15 min	20 min	25 min	30 min	Total (min)	% Inhibition
Control	12.55 ± 1.96	25.11 ± 3.43	20.11 ± 1.86	16.75 ± 3.06	12.14 ± 2.82	9.85 ± 2.11	89.77 ± 11.1	—
1 ml/kg	6.16 ± 1.03	9.5 ± 1.38	7.18 ± 0.74	5.5 ± 0.7	4.5 ± 1.02	2 ± 0.71	34.83 ± 4.49*	61.2
5 ml/kg	3.83 ± 0.74	11.83 ± 0.64	8.33 ± 0.49	7.16 ± 0.58	4.33 ± 0.52	2.07 ± 0.82	40 ± 2.48*	55.4
10 mm/kg	6.75 ± 1.34	15.16 ± 1.22	11.91 ± 1.59	10.77 ± 0.88	7.88 ± 1.56	4.62 ± 1.05	50.61 ± 6.18*	43.8

Values are mean ± S.E.M. (n=10), *p<0.05 significantly different from control (Student's t-test).

(Dr. HMIIPHS) and were housed in groups of 6 per cage for seven days prior to experimentation with free access to food and tap water ad libitum and kept on a 12h light/dark cycle. Each experimental group consisted of six animals. All animals were housed in an air-conditioned room at 23±1°C during the quarantine period.

Effect of oil on the Writhing Response Induced by Acetic Acid

The writhing test described by Koster *et al.* (1959) was adopted. A total of 24 Mice divided into four groups (n=6) were used and treated as follows; group I received normal saline (10ml/kg) p.o., group 2-4 received 1, 5, and 10ml/kg p.o. of the herbal oil respectively. 30 min later, 10ml/kg of 0.7% aqueous solution of acetic acid were given to all mice. Each mouse was placed in a transparent observational cage and the abdominal constrictions resulting from intraperitoneal injection of Acetic Acid consisting of the contraction of abdominal muscle together with a stretching of hind limbs, were counted according to procedure described by Santos *et al.* (1994), Correa *et al.* (1996), Bersa *et al.* (1996). Result were presented as percent inhibition of analgesia, calculated as the reduction in the number of writhes between control animals and those pretreated with either dose of oil.

Formalin induced nociception:

This test was carried out in four group of mice (n=6) pretreated with either saline or the oil (1, 5 and 10ml/kg) p.o. Twenty microliters of 1% Formaldehyde was injected into the sub-plantar surface of right hand paw of mice (Hunskar *et al.*, 1985; Tjolsen *et al.*, 1992). The time in seconds spent in licking and biting responses of the injected paw was taken as an indicator of pain response in the mice. Responses were measured for 0-5 min (first phase) and 15-30 min (second phase) after formalin injection. Oil (1, 5 and 10ml/kg) was administered 30 min before formalin injection. Control group was treated with Saline (10 ml/kg).

Carrageenin-induced paw edema in rats:

The paw edema/Inflammation was induced by carrageenin (0.1 ml, 1% w/v in normal saline) into sub-plantar tissue of right hind paw (Winter *et al.*, 1962). The linear paw circumference was measured at hourly interval for 6 h (Bamgbose and Noamesi, 1981).

Herbal oil (1 and 10 ml/kg) and saline (10 ml/kg) were administered orally 1h before induction of inflammation.

Acute toxicity:

Mice were administered orally with the herbal oil (1, 5 and 10 ml/kg) and the incidence of Mortality was noted up to 24 h in each group.

Table 3
Effect of PHOE on formalin induced nociception in mice

Group	Score of Pain		Score of Pain	
	0-5 min	% inhibition	15-30 min	% inhibition
Control	1.67 ± 0.19	—	4.002 ± 1.28	—
1 ml/kg	1.12 ± 0.03*	32.9	1.09 ± 0.48*	72
5 ml/kg	0.94 ± 0.18*	43.7	0.8 ± 0.29*	80
10 ml/kg	0.87 ± 0.19*	47.9	0.74 ± 0.015*	81.5

Values are mean ± S.E.M. (n=6). *P<0.05 significantly different from control (Student's t-test).

Table-4
Effect of PHOE on carrageenan-induced rat paw edema

Group	Carregeenin-induced rat paw edema	
	Increase in paw diameter	% inhibition of paw edema
Control	0.54 ± 0.08	—
1 ml/kg	0.35 ± 0.005	35.2
10 ml/kg	0.2 ± 0.07	62.9

Values are mean ± S.E.M. (n=6). *P<0.05 significantly different from control (Student's t-test).

STATISTICAL ANALYSIS

Results are presented as means ± S.E.M. Student's t-test was applied to determine a significant between the control group and experimental groups in all the experiments. P<0.05 was considered significant.

RESULT

Acetic Acid – induced writhing in mice:

The intraperitoneal injection of 0.6% (v/v) solution of acetic acid in mice induced a writhing response. The PHOE exhibited marked reduction in the number of abdominal constriction between 0 and 30 min (Table 2). This reduction is more pronounced at low dose than higher dose. At 10, 5 and 1 ml/kg it cause 43, 55, 61% inhibition respectively (Table 2).

Formalin-induced paw licking in mice:

PHOE evaluated against formalin induced pain in mice at 1, 5 and 10 ml/kg and the effects in early and late phases of formalin test

are shown in Table 3. PHOE was active in both early and late phases of the formalin test showing a pain inhibition of 47, 43 and 32% of the early phase (0-5 min) and 81, 80, 72% of the late phase for 10, 5 and 1 ml/kg of the oil, respectively (Table 3). Higher pain inhibition was, however, observed in the late phase.

Carrageenin-induced edema:

The results of animal experiments are shown in Table 4. In the acute inflammatory model, the oil in the dose of 10 and 1 ml/kg p.o. showed pronounced dose dependent inhibition of paw edema that was 63 and 35% respectively (Table 4).

Toxicity Studies:

PHOE was well tolerated up to a dose of 10ml/kg, through oral route. No mortality was observed during 24h period. In the initial pharmacological screening, it was observed that the dose produced palpebral ptosis a reduction in locomotion (non quantified observation).

DISCUSSION

This study is aimed to evaluate the scientific basis and to clarify the traditional use of PHOE against pain and inflammation as painkiller in folk medicine. Acetic acid induced abdominal constriction is used method for evaluation of peripheral analgesic effect (Gene *et al.*, 1998). The results showed that PHOE exhibited antinociceptive potency against chemical induced nociception (Pain) in mice. It significantly inhibit the abdominal constriction induced by acetic acid in mice. Acetic acid cause inflammatory pain by inducing capillary permeability (Amico--Roxas, 1984) and cause an increase in peritoneal fluids of PGE2 and PGF2 α which increases the sensitivity of nociceptors and perception of pain (Daeaedt *et al.*, 1980; Bentley *et al.*, 1983). Therefore, the observed effects in this study suggest that at all doses, PHOE analgesic effect could be peripherally mediated, might have occurred via prostaglandin synthesis inhibition.

Injection of formalin in sub-planter surface of right hand paw of mice produces a distinct biphasic pain, termed early and late phase. Shibata *et al.* (1989) reported that drugs that act primarily on the central nervous system inhibited both phases equally while peripherally acting drugs inhibit the late phase. In the present study PHOE inhibit both phases of formalin test indicated that central and peripheral both mechanisms are involved. However, higher inhibition was observed in the late phase strengthening the peripheral analgesic action of PHOE in addition to central mechanism of analgesia.

Earlier studies had indicated that carrageenin induced inflammation is the standard experimental model for acute inflammation, is known to be a peripheral process (Ahmadiani *et al.*, 2000). Effect of the PHOE on carrageenin-induced paw edema was most pronounced and results showed that suppress the exudates inflammation, therefore, suggest that it possesses anti-inflammatory activity.

In our opinion, all the constituents of PHOE are used in Azerbaijani traditional medicine, Chinese pharmacopoeia and in American conventional medicine in relieving pain and inflammation. All the effects independently produced by various ingredients in PHOE are producing a total beneficial effect.

The present results indicate that PHOE used traditionally for the treatment of pain and inflammation have ability to reduce the number of writhes, inhibit both phase of formalin-induced pain and suppression of carrageenin induce edema. Apparently analgesic effect of PHOE is probably mediated via inhibition of prostaglandin synthesis, suggesting more of a peripheral activity and confirms its analgesic and anti-inflammatory properties that validates the claim and folkloric use of PHOE in Unani system of medicine for the management of pain.

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